



DiaMedica Therapeutics Inc.
Phase 2/3 Adaptive Design, Randomized Double-blind Placebo-controlled Study to Evaluate
the Safety and Efficacy of DM199 for the Treatment of Acute Ischemic Stroke
(ReMEDy2 Trial)

Protocol Number: DM199-2021-001

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NCT Number:	05065216
Name of Investigational Product:	DM199 (rinvecalinase alfa)
Phase of Development:	2/3
Indication:	Treatment of Acute Ischemic Stroke
Sponsor:	DiaMedica Therapeutics Inc. 301 Carlson Parkway, Suite 210 Minneapolis, MN 55305 USA Tel: (763) 312-6064

National Principal Investigator:	Scott Kasner, M.D. University of Pennsylvania
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Protocol Title: Phase 2/3 Adaptive Design, Randomized Double-blind Placebo-controlled Study to Evaluate the Safety and Efficacy of DM199 for the Treatment of Acute Ischemic Stroke (ReMEDy2 Trial)
Protocol Number: DM199-2021-001

This study will be conducted in compliance with the clinical study protocol (and amendments), International Council for Harmonisation (ICH) guidelines for current Good Clinical Practice (GCP), World Medical Association Declaration of Helsinki, and applicable regulatory requirements.

Sponsor Signatory

Printed Name: Lorianne Masuoka, MD
Title: Chief Medical Officer

Signed by Lorianne Masuoka, MD
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Date (DD-MM-YYYY)

I approve this document
27-Aug-2024 | 2:35:11

INVESTIGATOR SIGNATURE PAGE

Protocol Title: Phase 2/3 Adaptive Design, Randomized Double-blind Placebo-controlled Study to Evaluate the Safety and Efficacy of DM199 for the Treatment of Acute Ischemic Stroke (ReMEDy2 Trial)

Protocol Number: DM199-2021-001

1. I, the undersigned, have reviewed this protocol (and amendments), including appendices, and I will conduct the study as described in compliance with this protocol (and amendments), Good Clinical Practice (GCP), and relevant International Council for Harmonisation (ICH) guidelines.
2. I am thoroughly familiar with the appropriate use of the study drug, as described in this protocol and any other information provided by DiaMedica Therapeutics Inc. including, but not limited to, the current Investigator's Brochure.
3. Once the protocol has been approved by the Institutional Review Board (IRB) /Independent Ethics Committee (IEC), I will not modify this protocol without obtaining prior approval of DiaMedica Therapeutics Inc. and of the IRB/IEC. I will submit the protocol amendments and/or any informed consent form modifications to DiaMedica Therapeutics Inc. and the IRB/IEC, and approval will be obtained before any amendments are implemented.
4. I will ensure that all persons or parties assisting me with the study are adequately qualified and informed about the DiaMedica Therapeutics Inc. study drug and of their delegated study-related duties and functions as described in the protocol.
5. I will ensure that source documents and trial records that include all pertinent observations on each of the site's trial participants will be attributable, legible, contemporaneous, original, accurate, and complete.
6. I understand that all information obtained during the conduct of the study with regard to the participants' state of health will be regarded as confidential. No participants' names will be disclosed. All participants will be identified by assigned numbers on all case report forms, laboratory samples, or source documents forwarded to the Sponsor. Clinical information may be reviewed by the Sponsor or its agents or regulatory agencies. Agreement must be obtained from the participant before disclosure of participant information to a third party.
7. Information developed in this clinical study may be disclosed by DiaMedica Therapeutics Inc. to other clinical Investigators, regulatory agencies, or other health authority or government agencies as required.

Investigator Name (Print)

Investigator Signature Date (DD-MM-YYYY)

Title

Institution

TABLE OF CONTENTS

1.	PROTOCOL SUMMARY.....	9
1.1.	Synopsis.....	9
1.2.	Study Design.....	16
1.3.	Schedule of Activities (SoA).....	17
2.	LIST OF ABBREVIATIONS.....	19
3.	INTRODUCTION	21
3.1.	Study Rationale.....	21
3.1.1.	Dose Rationale.....	21
3.2.	Background on Acute Ischemic Stroke	23
3.3.	Background on DM199 and Mode of Action	23
3.3.1.	Pharmacological Class.....	24
3.3.2.	Nonclinical Studies.....	24
3.3.3.	Clinical Studies.....	25
4.	STUDY OBJECTIVES AND ENDPOINTS.....	31
4.1.	Study Objectives.....	31
4.2.	Study Endpoints.....	31
4.2.1.	Primary Outcome Measures	31
4.2.2.	Secondary Outcome Measures	32
4.2.3.	Pharmacokinetics	32
4.2.4.	Immunogenicity	32
4.2.5.	Exploratory Endpoints	32
5.	DESCRIPTION OF OVERALL STUDY DESIGN AND PLAN	33
5.1.	Revised IV Dose	33
5.2.	Statistical Considerations.....	34
5.3.	Optional PK Sub-Study	34
5.4.	End of Study	34
6.	STUDY POPULATION.....	35
6.1.	Inclusion Criteria	35
6.2.	Exclusion Criteria	35
6.3.	Study Participant Flow Diagram	38
6.4.	Screen Failures.....	38

6.5.	Rescreening.....	38
7.	STUDY INTERVENTION	40
7.1.	DM199 Investigational Medicinal Product	40
7.2.	Placebo.....	40
7.3.	Study Supplies provided by Sponsor.....	40
7.4.	Study Drug Accountability	40
7.5.	Study Drug Administration.....	41
7.5.1.	Intravenous Treatment Administration.....	41
7.5.2.	Subcutaneous Treatment Administration	44
7.6.	Measures to Minimize Bias, Study Treatment Assignment and Masking.....	44
7.6.1.	Method of Study Treatment Assignment.....	44
7.6.2.	Unmasking	44
7.7.	Study Intervention and Assessment Compliance	45
7.8.	Concomitant Medications and Procedures	47
7.8.1.	Prohibited Therapies	47
7.8.2.	Procedures.....	47
8.	STUDY TREATMENT DELAYS, DISCONTINUATION, AND EARLY WITHDRAWALS	49
8.1.	Dose Reduction.....	49
8.1.1.	Dose Delays and Missed Doses	49
8.1.2.	Participant Discontinuation from Study Treatment.....	49
8.1.3.	Participant Withdrawal from Study	50
8.1.4.	Lost to Follow-Up.....	51
9.	STUDY PROCEDURES	52
9.1.	Descriptions of Assessments	52
9.1.1.	Medical History/Demographics	52
9.1.2.	Stroke Mechanism Classification	52
9.1.3.	Efficacy Assessments	54
9.1.3.1.	Stroke Scale Certification	54
9.1.3.2.	Modified Rankin Scale (mRS).....	54
9.1.3.3.	NIH Stroke Scale	54
9.1.3.4.	Barthel Index	54
9.1.3.5.	Stroke Recurrence.....	55

9.1.4.	Pharmacokinetics	55
9.1.5.	Safety Assessments.....	55
9.1.5.1.	Physical Examination	56
9.1.5.2.	Vital Signs	56
9.1.5.3.	Electrocardiograms	57
9.1.5.4.	Laboratory Assessments	58
9.1.5.5.	Unscheduled Visit.....	59
9.1.6.	Immunogenicity	59
9.2.	Adverse Events and Serious Adverse Events	59
9.2.1.	Definition of Adverse Events	59
9.2.2.	Definition of Serious Adverse Events	60
9.2.3.	Adverse Events of Special Interest Definition and Reporting.....	60
9.2.4.	Classification of an Adverse Event.....	63
9.2.4.1.	Assessment of Severity.....	63
9.2.4.2.	Relationship to Study Treatment	64
9.2.4.3.	Expectedness.....	64
9.2.5.	Adverse Event Collection and Reporting	64
9.2.5.1.	Adverse Event Collection	64
9.2.5.2.	Adverse Event Reporting.....	65
9.2.5.3.	Serious Adverse Event Reporting.....	66
9.2.5.4.	Adverse Event Follow-Up	66
9.2.5.5.	Pregnancy Reporting	67
9.2.6.	Suspected Unexpected Serious Adverse Reactions	67
9.2.6.1.	Overdose	68
10.	UNANTICIPATED PROBLEMS	69
10.1.	Unanticipated Problem Reporting	69
11.	STATISTICAL ANALYSIS	70
11.1.	Determination of Sample Size	70
11.2.	Analysis Populations	70
11.2.1.	Intent-to-treat Population.....	71
11.2.2.	Modified Intent-to-treat Population	71
11.2.3.	Per-Protocol Population.....	71
11.2.4.	Safety Population.....	71

11.3.	Interim Analysis.....	71
11.4.	Baseline Summaries.....	72
11.5.	Efficacy Endpoints and Analyses	72
11.5.1.	Primary Efficacy Endpoint: Modified Rankin Scale, Dichotomous	72
11.5.2.	Secondary Efficacy Endpoints and Analyses	72
11.5.2.1.	Modified Rankin Scale – Shift	73
11.5.2.2.	Modified Rankin Scale – Dichotomous.....	73
11.5.2.3.	Mortality	73
11.5.2.4.	National Institutes of Health Stroke Score	73
11.5.2.5.	Barthel Index Score	73
11.5.2.6.	Recurrent AIS	73
11.6.	Pharmacokinetics	74
11.7.	Immunogenicity	74
11.8.	Exploratory Endpoints	74
11.9.	Safety Analysis	74
11.9.1.	Adverse Events	74
11.9.2.	Injection Site Reactions	74
11.9.3.	Physical Examination Findings	75
11.9.4.	Vital Signs Results.....	75
11.9.5.	Hematology and Chemistry	75
11.9.6.	12-Lead ECG	75
11.9.7.	Concomitant Medications	75
11.10.	Subgroup Analyses	75
11.11.	Handling of Missing Data.....	75
12.	STUDY MANAGEMENT	76
12.1.	Approval and Consent	76
12.1.1.	Regulatory Guidelines	76
12.1.2.	Institutional Review Board / Ethics Committee	76
12.1.3.	Informed Consent	76
12.2.	Study Discontinuation and Closure	77
12.3.	Confidentiality and Privacy	78
12.4.	Steering Committee	78
12.5.	Data and Safety Monitoring Board.....	78

12.6.	Conflict of Interest Policy.....	79
12.7.	Monitoring	79
12.7.1.	Quality Control and Quality Assurance.....	80
12.8.	Data Collection	80
12.8.1.	Data Management.....	80
12.8.2.	Source Documents	80
12.8.3.	Protocol Deviations	81
12.8.4.	Record Retention	81
12.8.5.	Storage of Samples	81
12.9.	Protocol Amendment.....	82
12.10.	Publication Policy/Disclosure of Data.....	82
13.	REFERENCES	83
APPENDIX 1.	MODIFIED RANKIN SCALE	85
APPENDIX 2.	NIH STROKE SCALE	86
APPENDIX 3.	NORMAL RANGES (VITAL SIGNS AND ECG).....	94
APPENDIX 4.	BARINDEX	95

LIST OF TABLES

Table 1:	Schedule of Activities (SoA).....	17
Table 2:	NIHSS Score and Prior AIS Treatment in ReMEDy1.....	28
Table 3:	Management of DM199 or placebo IV Administration	42
Table 4:	Causative Classification for Ischemic Stroke	53
Table 5:	Laboratory Assessments	58
Table 6:	AESI Reporting from the CTCAE v5.0.....	62
Table 7:	CTCAE Grading	63

LIST OF FIGURES

Figure 1:	Study Design.....	16
Figure 2:	DM199 NIHSS Outcomes in ReMEDy1.....	29
Figure 3:	ReMEDy1 Study – Clinical Outcomes NIHSS and mRS ≤ 1 and Deaths	30
Figure 4:	Study Participant Flow Diagram	38

1. PROTOCOL SUMMARY

1.1. Synopsis

Title of Study:	Phase 2/3 Adaptive Design, Randomized Double-blind Placebo-controlled Study to Evaluate the Safety and Efficacy of DM199 for the Treatment of Acute Ischemic Stroke (ReMEDy2 Trial)
Protocol Number:	DM199-2021-001
Investigators/Study Sites:	Approximately 100 sites; this number may increase depending on the results of the interim analysis
Phase of Development:	2/3
Objectives:	The objective of this study is to evaluate the safety and efficacy of DM199 (rinvecalinase alfa) in participants with moderate stroke severity, who present within 24 hours of Acute Ischemic Stroke (AIS) onset due to small and medium vessel occlusions. This study focuses on participants who have limited treatment options. Participants who have or will receive mechanical thrombectomy (MT) are not eligible for participation. Additionally, participants who have received fibrinolytics are excluded unless they experience a persistent neurological deficit of moderate severity six or more hours after fibrinolytic treatment. Participants considered for this trial should not be denied the use of standard of care (SoC) AIS therapies, such as fibrinolytics or MT, when appropriate.
Study Endpoints:	Primary Outcome Measures: <u>Efficacy</u> - Stroke recovery as defined by participants with excellent functional outcomes at Day 90 as assessed via the modified Rankin Score (mRS [dichotomized]), mRS scores of 0 or 1 represent responders, scale range of 0–6. <u>Safety</u> - Safety as assessed by: - Incidence, severity, and causality of adverse events (AEs) and serious adverse events (SAEs). - Incidence, severity, and causality of adverse events of special interest (AESI). - Tolerability (incidence and severity of injection site adverse reactions). - Change from Baseline in physical examination (PE) at Day 21 and Day 90. - Change from Baseline in vital sign measurements (resting heart rate (HR), systolic/diastolic blood pressure (BP), respiratory rate (RR), and body temperature (BT)) at Days 1, 4, 21, 30 and 90. - Change from Baseline in hematology and chemistry parameters at Days 4, 30, and 90. - Change from baseline in 12-lead electrocardiogram (ECG) at Days 1 and 90 as determined by the Investigator. Secondary Outcome Measures: - Assessment of effect on disability across the full spectrum of AIS by examining the distribution of mRS (shift) scores (scale range = 0 to 6) at Day 90.

	2. Proportion of participants achieving independent function (able to look after their own affairs without assistance) with or without minor disability at Day 90 assessed as mRS 0-2 (dichotomized) mRS scores of 0, 1 or 2 represent responders, scale range of 0–6. 3. Mortality rate as defined by event rate (%) for mortality over 90 days. 4. Proportion of participants achieving an excellent neurological outcome defined by NIHSS = 0-1 (dichotomized) (NIHSS scores of 0 or 1, scale range 0 to 42) at Day 90. 5. Proportion of participants achieving an excellent functional independence in activities of daily living defined by Barthel Index score (dichotomized) greater than or equal to 95 (scale range 0 to 100) at Day 90. 6. Recurrent AIS as defined by proportion of participants who experience a recurrent AIS by Day 90 as assessed by a new, persistent neurological deficit attributable to cerebrovascular ischemia. Imaging findings, if available, should support the diagnosis. Pharmacokinetics: Plasma concentration of KLK1 (Baseline and Day 1, Day 2, Day 21, and Day 90). Immunogenicity: Plasma DM199 anti-drug antibodies (ADA) (Baseline and Days 2, 21, and 90). Exploratory Endpoints: 1. Estimated Glomerular Filtration Rate (eGFR) (Baseline and Day 4, Day 21, and Day 90). 2. Urinary Albumin Creatine Ratio (UACR) (Baseline and Day 4, Day 21, and Day 90).
Study Design:	This is a Phase 2/3 Adaptive Design, Randomized, Double-blind, Placebo-controlled Study to Evaluate the Safety and Efficacy of DM199 for the Treatment of Acute Ischemic Stroke (ReMEDy2 Trial). Participants with AIS will be randomized 1:1 to DM199 or placebo. DM199 will be administered as a single intravenous (IV) dose (0.5 µg/kg; not to exceed 50 µg) followed by subcutaneous (SC) doses at 2 hours (+10 hours) of the IV dose completion and then 2 times per week up to Day 21. The duration of each individual's participation in the study will be approximately 90 days from the time of initial treatment to completion of all study activities. Enrollment will be in a randomized 1:1 ratio (DM199 vs placebo) and double-blind manner. When 20 participants (including 11 participants enrolled post-hold) have completed IV dosing, the independent Data Safety Monitoring Board (DSMB) will review all unmasked safety data to determine if any risk mitigations are recommended. The DSMB will perform periodic safety data review to ensure that continuation of the study does not pose a health hazard to participants as outlined in a separate DSMB Charter. The DSMB will provide this information to the Institutional Review Board (IRB)/Independent Ethics Committees (IEC) to address any safety concerns, and offer direction at interim analysis (e.g., move forward with the trial, update the protocol, or stop the trial).

	Blood samples for the determination of KLK1 in plasma will be collected in a PK sub-study, at selected sites. Samples will be collected at 9 timepoints including Baseline and Day 1 (6 timepoints), Day 2, Day 21, and Day 90. Sparse PK samplings will be collected at all other sites at Baseline and Day 1 (2 timepoints), Day 2, Day 21, and Day 90 for a total of 5 timepoints. Blood samples will also be collected for the determination of plasma DM199 ADA at baseline and on Days 2, 21, and 90. A formal interim analysis will be conducted after 144 participants complete their Day 90 assessment (Part A). The purposes of this interim analysis are to assess safety, allow early stopping of the study for futility, or continuing the study with a revised sample size up to a maximum of 728 total participants (Part B).
Selection of Participants:	Inclusion Criteria: - Participant is between 18 and 90 years of age inclusive. - Participant weight is 40 kg to 166 kg inclusive. - Participant to be randomized and treatment initiated within 24 hours of last known normal/AIS stroke onset. - Participant has NIHSS ≥ 5 and ≤ 15 at approximately the time of randomization. This criterion also applies to participants who meet the following conditions: - The participant initially presents with an NIHSS score below 5 but clinically worsens, including cases of progressing stroke / stroke-in-evolution, resulting in a subsequent persistent NIHSS score of ≥ 5 and ≤ 15; and - Participant meets all other inclusion and exclusion criteria, including repeat brain imaging to rule out hemorrhagic transformation - Participant had a pre-morbid mRS score of 0 to 1 (mRS score prior to AIS) as stated by participant or participant's representative. - If participant has received fibrinolytic treatment for AIS within 4.5 hours of last known normal/AIS stroke onset and at least 6 hours after completing fibrinolytic treatment, and the participant meets all the following criteria: - Participant's initial NIHSS score prior to fibrinolytics was ≤ 15; and - At least six hours after fibrinolytics, the participant has NIHSS score of ≥ 5 and ≤ 15 with a persistent deficit; and - The participant's NIHSS score showed less than a 4-point improvement, or worsened, after receiving fibrinolytics; and - Participant meets all other inclusion and exclusion criteria including repeat brain imaging to rule out hemorrhagic transformation. - Participant and/or legally authorized representative is able to provide informed consent. - Participant is willing and able to comply with the study protocol, in the Investigator's judgment.

	Exclusion Criteria: 1. At screening, or with repeat imaging (see Inclusion 4 and 6), participant has imaging confirmed hemorrhagic stroke 2. Participant has image findings with symptomatic large vessel occlusion at one or more of the following locations: Intracranial carotid I/T/L or M1 segment MCA, vertebral or basilar artery (BA). 3. Participant has large core of established infarction defined as ASPECTS 0-5 4. Participant has or will receive MT for their current AIS. 5. Participant has suspected or confirmed extracranial arterial dissection 6. Participant has imaging findings and/or symptoms consistent with a brain stem or cerebellar stroke. Posterior cerebral artery strokes without any associated brain stem or cerebellar involvement are allowable. 7. Participant has any recorded SBP < 100 mm HG or MAP <65 mm Hg; MAP = DBP + [1/3 (SBP – DBP)] (measured with noninvasive BP cuff type monitor) after stroke symptom onset and prior to randomization. 8. Participant is currently prescribed angiotensin-converting enzyme inhibitor (ACEi) and is unable or unwilling to convert to another antihypertensive pharmacological treatment through Day 29 ±1 day (8 days after last treatment). 9. Participant is currently prescribed an ACEi, and the last dose of the ACE inhibitor medication is reported to have been taken < 24 hours before start of IV study drug infusion as stated by participant or participant’s representative. 10. Participant has a history of clinically significant allergic reactions such as angioedema or anaphylaxis requiring hospitalization. 11. Participant has a diagnosis or suspected diagnosis of hereditary angioedema (HAE) or is taking or prescribed medications commonly used as prophylaxis/treatment of HAE, such as C1-esterase inhibitors (Cinryze, Berinert, Ruconest, Haegarda), Danazol, kallikrein inhibitors (Ecallantide, Berotralstat, Lanadelumab), Bradykinin B2 Receptor Antagonists (Icatibant), or other medication designed to influence the kallikrein-kinin system. 12. Life expectancy estimated at ≤1 year prior to enrollment. 13. Participant has clinical evidence of an active infection at the time of enrollment requiring parenteral treatment or hospitalization to monitor or manage the infection.
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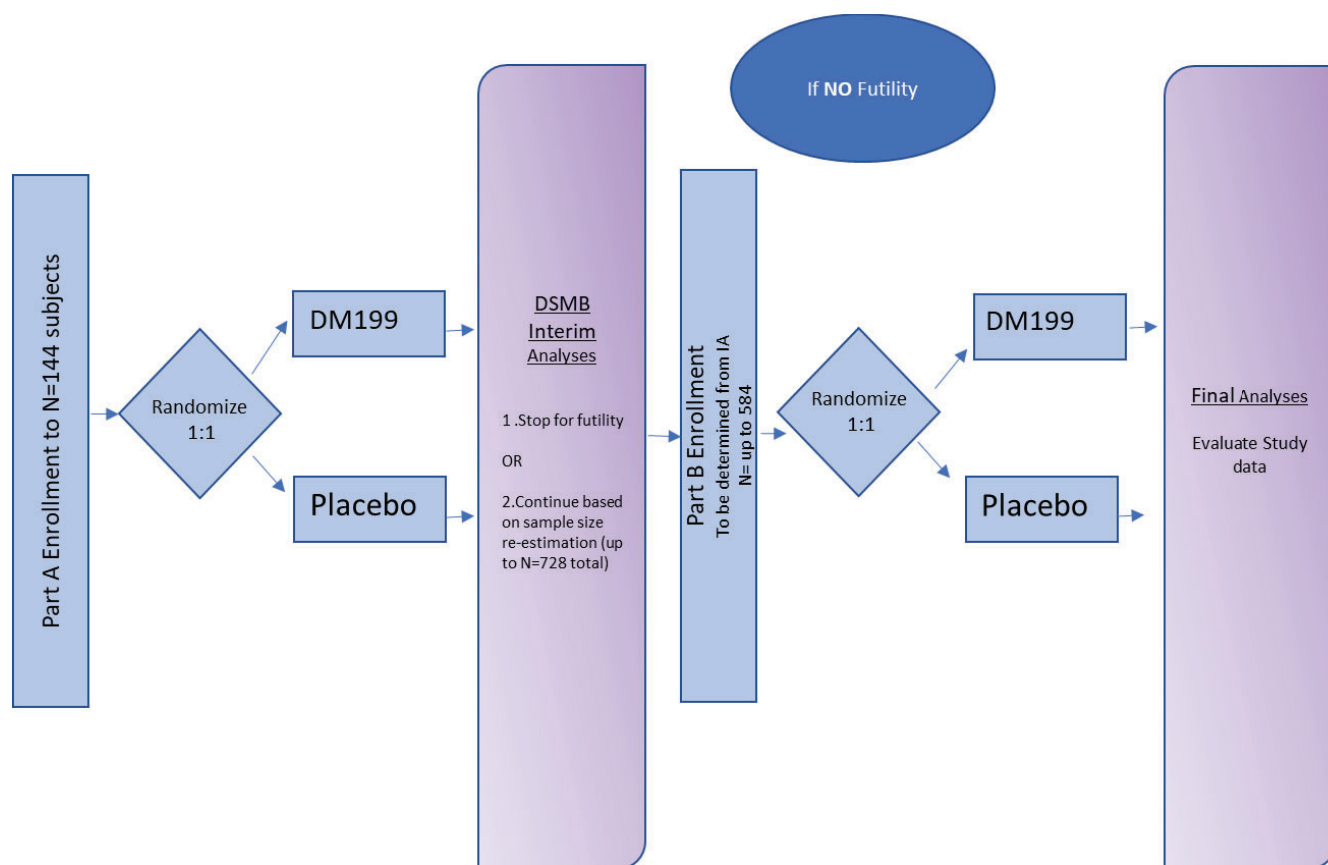
	<i>NOTE: Treatment of uncomplicated infections with oral antibiotics would not be an exclusion (example treatment of an uncomplicated urinary tract infection or sinus infections with oral antibiotics would not be an exclusion).</i> 14. Participant has known alpha 1-antitrypsin deficiency (α1-antitrypsin deficiency). 15. Participant is pregnant or nursing. <i>NOTE: Participants who agree to stop nursing may be considered for inclusion at the discretion of the Investigator.</i> 16. Participants of child-bearing potential must agree to use medically acceptable contraceptive measures to prevent pregnancy. All participants of childbearing potential (defined as sexually mature participants who have had menses within the preceding 24 months and have not undergone permanent sterilization methods such as hysterectomy, bilateral oophorectomy, bilateral salpingectomy, etc.) must have a negative serum pregnancy test performed locally at screening. Participants of childbearing potential must agree not to attempt to become pregnant or undergo in vitro fertilization. If participating in sexual activity that could lead to pregnancy, participants must use 2 reliable methods (1 per partner is acceptable) of contraception simultaneously while receiving protocol-specified medication and during the study follow-up period. Participants participating in sexual activity must agree to use, or for their partner to use highly effective birth control methods (those with a failure rate of less than 1% per year when used consistently and correctly) until they have completed the study (after the Day 90 visit). Such methods include: • Combined (estrogen and progesterone containing) hormonal oral, intravaginal, or transdermal contraception associated with the inhibition of ovulation • Progesterone-only oral, injectable, or implantable hormonal contraception associated with the inhibition of ovulation • Intrauterine device (IUD) • Intrauterine hormone-releasing system (IUS) • Bilateral tubal occlusion • Vasectomized partner • Sexual abstinence Participants who are not of reproductive potential (who have been postmenopausal for more than 24 consecutive months or have undergone hysterectomy, bilateral oophorectomy) are not required to use contraception. Participants are prohibited from sperm donation. <i>NOTE: A negative serum pregnancy test will be documented during screening if a participant is of child-bearing potential.</i> 17. Participant is currently participating in or has participated in a study using an investigational device or drug or received an
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	investigational drug or investigational use of a licensed drug within 30 days prior to screening. 18. Participant does not have sufficient venous access for infusion of study treatment or blood sampling. 19. Participant is unable or unwilling to comply with protocol requirements, including assessments, tests, and follow-up visits. 20. Participant has any other medical condition which in the opinion of the Investigator will make participation medically unsafe or interfere with the study results.
Planned Sample Size:	It is assumed that the efficacy responder rates (i.e., proportions of participants with mRS scores of 0 or 1) at Day 90 are 15% and 29% for placebo and DM199, respectively. With these assumptions, 358 participants (179 participants per treatment group) would be required to achieve 90% power at the nominal one-sided significance level of $\alpha=2.5\%$. With an estimated rate for early study discontinuation of less than 2%, a total of 364 participants (182 participants per treatment group) are planned to be randomized in 1:1 ratio to receive DM199 or placebo. An interim analysis is planned at the end of Part A after 144 participants complete their Day 90 assessments (approximately 40% of the pre-planned, total sample size for Parts A and B). The sample size will be re-estimated at the end of Part A to determine if the sample size in Part B should be revised up to a maximum of 584 for a total possible maximum of 728 participants.
Investigational Therapy:	Active Substance: DM199 (rinvecalinase alfa) Activity: Recombinant form of the endogenous serine protease protein, human tissue kallikrein-1 (rhKLK1) Indication: Treatment of Acute Ischemic Stroke within 24 hours of estimated onset, where participant did not receive or did not respond to fibrinolytics and/or Mechanical Thrombectomy was not indicated or medically appropriate. Dose Regimen: DM199 or placebo will be administered on: - Day 1 by a single IV infusion of 0.5 µg/kg of DM199 (not to exceed 50 µg) in normal saline or placebo (50 mL total) - Followed by first SC injection of 3 µg/kg of DM199 or placebo at 2 hours (+10 hours) of IV infusion completion - Then SC injection of 3 µg/kg of DM199 or placebo 2x per week until Day 21 (total of 7 SC doses)
Reference Therapy:	Matching placebo containing all manufactured ingredients/excipients as the study drug but without the active pharmaceutical ingredient for intravenous infusion and subcutaneous injection.
Treatment Duration:	This is a 90-day study with study drug administration Day 1 through Day 21.
Pharmacokinetics:	Blood samples for the determination of KLK1 in plasma will be collected at Baseline and Day 1 (2 timepoints), Day 2, Day 21, and Day 90 for a total of 5 timepoints.

	Blood samples for participants that agree to participate in the optional PK sub-study will have collection timepoints at Baseline and Day 1 (6 timepoints), Day 2, Day 21, and Day 90 for a total of 9 timepoints.
Immunogenicity:	Plasma DM199 anti-drug antibodies (ADA) (Baseline, Day 2, Day 21, and Day 90).
Statistical Methods and Planned Analyses:	The statistical analysis plan (SAP) will provide further details regarding the definition of analysis variables and analysis methodology and will serve as a complement to the protocol and supersede it in case of differences in definition of analysis variables and analysis methodology. The statistical evaluation will be performed using SAS® software version 9.4 or higher (SAS Institute, Cary, NC). All data collected on the eCRFs and results from external data, e.g., PK and ADA data, will be presented in the data listings. In general, the efficacy, safety, and PK endpoints will be summarized descriptively by treatment group and timepoints or other variables, as appropriate. A formal interim analysis will be conducted after 144 participants complete their Day 90 assessment in Part A. The purposes of this interim analysis are to assess safety, allow early stopping of the study for futility, or continuing the study with a revised final sample size of up to 728. The futility rule will be non-binding. All statistical comparisons will be made using a one-sided test at the $\alpha = 0.025$ significance level unless specifically stated otherwise. Null hypotheses will be no treatment difference or active treatment has worse outcomes than placebo and alternative hypotheses will be one-sided where the active group has better outcomes than placebo. The primary efficacy endpoint and analysis is based on AIS Recovery, as defined by participants with excellent functional outcomes at Day 90 as assessed via the mRS (dichotomized) where mRS scores of 0 or 1 represent responders. The comparison between the treatment groups using the Cochran Mantel Haenszel test, stratified by NIHSS and prior use of fibrinolytics randomization factor, will be performed as the primary analysis. Further details on the primary analysis as well as secondary and exploratory analyses can be found in Section 13 as well as the SAP.

1.2. Study Design

Figure 1: Study Design



1.3. Schedule of Activities (SoA)

Please refer to Section 7.5 Study Drug Administration and Section 9 Study Procedures for complete description.

Table 1: Schedule of Activities (SoA)

Timepoint	Screening	Day 1						Active Treatment Period (continued)				Follow-Up Period		Early Withdrawal*	Unscheduled Visit*
		Baseline (Pre-Dose)	Study Drug IV Infusion	During IV Infusion	Post-IV Infusion	Study Drug SC Dose	Post-SC Dose	Day 2	Day 4 (+/- 1 day)	Days 7, 11, 14, 18 (+/- 1 day)	Day 21 (+/- 1 day)	Day 30 (-3 days to +7 days)	Day 90 + 14 days		
Informed Consent	X														
Medical History/Demographics	X														
Previous and Concomitant Medications	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Vital Signs (BP and HR ONLY)	X ¹		X ⁶	X ⁶	X ⁶										
Vital Signs (RR, HR, BP, and BT) ¹⁴		X ⁵			X ⁷		X ⁸		X		X		X	X	X
Pre-Morbid Modified Rankin Scale (mRS)	X														
NIH Stroke Scale (NIHSS)	X ²											X	X	X	X
Serum Pregnancy Test WCBP	X ³														
Inclusion/Exclusion Criteria Review	X														
Randomization	X ⁴														
Physical Examination		X									X		X	X	X
Weight, Height, and BMI calculation		X									X ¹⁷		X ¹⁷	X ¹⁷	X ¹⁷
12-Lead ECG		X			X ¹⁵		X ¹⁶						X	X	X
Central Clinical Laboratory	X								X ¹³		X ¹³		X	X	X
Blood Sampling for KLK1 ADA		X ⁵						X ¹²			X ¹³		X	X	X
Optional PK Sub-Study for KLK1	X ⁹	X ⁹		X ¹⁰	X ¹⁰			X ¹²			X ¹³		X	X	X
PK Sampling for KLK1		X ⁵			X ¹¹			X ¹²			X ¹³		X	X	X
Modified Rankin Scale (mRS)		X										X	X	X	X
Barthel Index		X										X	X	X	X
Urine Pregnancy Test WCBP											X		X	X	X
Phone Follow-up/Telehealth Visit										X		X			
Home Health Visit									X ¹⁸	X ¹⁸					
Stroke Mechanism Classification													X ¹⁹	X ¹⁹	
Study Drug Administration ²⁰			X			X ²¹			X	X	X				
Adverse Event Monitoring	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Abbreviations: BMI = body mass index, ECG = electrocardiogram, IV = intravenous, KLK1 = kallikrein, RR =Respiratory Rate, HR = Heart Rate, BP = Blood Pressure, BT = Body Temperature, SC = Subcutaneous ADA = Anti-drug Antibodies, PK = Pharmacokinetics, WCBP = Women of Child-Bearing Potential *Only complete Visit and/or Timepoint if applicable

Footnote Key:

1. ONLY BP needed for Inclusion/Exclusion.	9. For the optional PK (sub-study), DM199 Blood KLK1 PK at: A) with central clinical labs and B) at Pre-IV dose, as close to the IV start time as possible.	15. Perform ECG within 60 minutes of IV stop time.
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2. NIHSS must be performed immediately before Randomization.	10. For the optional PK sub-study, DM199 Blood sampling (KLK1 PK) at: A) 20min (+/- 5 min) of IV start time; B) IV Stop time (+10 min); C) 60 min (+/- 10 min) after IV stop time; and D) prior to Day 1 SC dose.	16. Perform ECG at 10 hours (+/- 2 hours) after first SC dose.
3. Must have a negative serum pregnancy test performed locally at screening.		
4. Randomization after all Inclusion/Exclusion Criteria met.	11. For PK sampling at all sites, DM199 blood sampling (KLK1 PK) at IV stop time (+10 mins).	17. Height and BMI not collected.
5. Collected as close to the IV start time as possible.	12. Collected a minimum of 12 hours after 1 st SC dose and Prior to Second SC dose.	18. Home Health Visit not needed if in-clinic visit.
6. ONLY BP and HR collected at A) start of IV infusion, B) 5 min (+/- 3 min) after IV start, C) every 10 min (+/- 5 min) until infusion complete, and D) every 15 min (+/- 5 min) for one hour after IV infusion complete.	13. Samples will be collected pre-SC dose.	19. Classification can be completed prior to Day 90; data collection expected by day 90.
7. 30 min (+/- 10 min) after IV infusion stop time.	14. Vital signs will be performed every 8 hours (+/- 2 hours) while participant is inpatient for initial AIS (per Institutional SoC).	20. SC doses must be a minimum of 24 hours apart.
8. 2 hours (+/- 30 min) post SC dose.		21. First SC injection should occur at 2 hours (+10 hours) of IV infusion completion.